AI SALES AGENT

Non-functional requirements

Quality attributes and measurable targets, with locked decisions separated from open assumptions.

Phase 1 MVP · 21 October 2026 · Prepared from the canonical documents in docs/

How to read the targets

Every row below is a documented requirement from `docs/00_product/01_PRODUCT_REQUIREMENTS` section 16. A target written in the table is an acceptance criterion, not a proposal. The last section of this document separates those targets from working assumptions that the founder has not locked.

16.1 Performance

ID	Requirement	Target or criterion	Verify
NFR-PERF-01	Webhook acknowledgement latency	p95 < 500 ms, p99 < 1 s	OB
NFR-PERF-02	Inbound message → first AI reply delivered	p95 < 30 s	OB
NFR-PERF-03	Read API responses (lists, detail)	p95 < 300 ms at 100 concurrent users	AT, OB
NFR-PERF-04	Inbox initial render with 50 conversations	< 2 s on a 3G-class connection	MT
NFR-PERF-04a	New inbound message visible in an open inbox	< 2 s after persist	OB
NFR-PERF-04b	Switching between cached conversations	< 200 ms	MT
NFR-PERF-04c	Take-over action confirmed server-side	< 500 ms	AT
NFR-PERF-04d	AI suggestion returned after request	< 5 s, with a visible pending state	OB
NFR-PERF-05	Product search tool execution	p95 < 400 ms	OB
NFR-PERF-06	Vector retrieval for a business with 5,000 chunks	p95 < 250 ms	AT
NFR-PERF-07	CSV import of 1,000 rows	completes < 60 s	AT

16.2 Scalability

ID	Requirement	Target or criterion	Verify
NFR-SCAL-01	Concurrent tenants supported on the Phase 1 ECS deployment	100 businesses	
NFR-SCAL-02	Sustained inbound message throughput	20 messages/second aggregate	
NFR-SCAL-03	Catalog size per business	10,000 variants without query degradation	
NFR-SCAL-04	Conversation history retained and queryable	1,000,000 messages per business	
NFR-SCAL-05	Application tier is stateless; shared state is in PostgreSQL, Redis, or S3	Three Fargate services from day one (`frontend`, `backend`, `worker`); scale by task count	
NFR-SCAL-06	Job workers scale independently of the API process	Separate ECS service, same NestJS image	

16.3 Availability and reliability

ID	Requirement	Target or criterion	Verify
NFR-REL-01	API and webhook availability	99.5% monthly (Phase 1)	
NFR-REL-02	No inbound customer message is lost once acknowledged	Zero tolerance; webhook acks only after durable persist	
NFR-REL-03	Job retry policy	3 attempts with exponential backoff, then dead-letter with alert	
NFR-REL-04	Duplicate side effects from retries	Zero; enforced by idempotency keys on messages, orders, and jobs	
NFR-REL-05	Recovery point objective (data loss)	≤ 5 minutes	
NFR-REL-06	Recovery time objective	≤ 4 hours	
NFR-REL-07	Graceful degradation when the LLM provider is unavailable	Fallback model, then queue; inbox continues; humans can reply; owner notified	
NFR-REL-08	Meta Graph / webhook unavailable	Outbound queued; inbound ack `200` only after durable persist	
NFR-REL-09	Tool execution failure	Customer is told of uncertainty; human escalation; failure logged	

16.4 Security and privacy

ID	Requirement	Target or criterion	Verify
NFR-SEC-01	All traffic over TLS 1.2+; no plaintext transport of credentials or tokens.		
NFR-SEC-02	Data encrypted at rest (database, object storage, backups).		
NFR-SEC-03	Provider tokens and API keys held in a managed secret store, never in source, environment files committed to VCS, or logs.		
NFR-SEC-04	Every tenant-scoped query is filtered by `business_id`; cross-tenant access is treated as a Sev-1 defect.		
NFR-SEC-05	Authorisation is enforced server-side on every endpoint against the permission matrix; UI hiding is not a control.		
NFR-SEC-06	Customer PII (phone, address) is redacted from application logs and from prompts sent to model providers where not required for the task.		
NFR-SEC-07	Webhook endpoints verify provider signatures before processing.		
NFR-SEC-08	Dependency and container vulnerability scanning runs in CI; critical findings block release.		
NFR-SEC-09	Rate limiting protects authentication, public webhooks, and per-tenant AI consumption.		
NFR-SEC-10	A business can request deletion of an end customer's personal data; the system supports targeted erasure.		

16.5 Internationalisation and localisation

ID	Requirement	Target or criterion	Verify
NFR-I18N-01	The AI must handle Tunisian Arabic, Modern Standard Arabic, Tunisian Arabizi and French, including code-switching within a single message.		

ID	Requirement	Target or criterion	Verify
NFR-I18N-02	Arabic content renders right-to-left correctly in the inbox, including mixed-direction text and embedded Latin product names.		
NFR-I18N-03	All merchant-facing UI strings are externalised for translation; no hard-coded display text.		
NFR-I18N-04	Currency, number, and date formatting follow the business locale and timezone.		
NFR-I18N-05	Phone number validation accepts Tunisian formats and is extensible to other MENA country codes.		

16.6 Usability and accessibility

ID	Requirement	Target or criterion	Verify
NFR-UX-01	A non-technical merchant can complete onboarding to AI activation unaided in under 20 minutes with a ready CSV.		
NFR-UX-02	The inbox is usable on mobile web at 375 px width; the merchant's primary device is a phone.		
NFR-UX-03	Every destructive action requires confirmation and is reversible or auditable.		
NFR-UX-04	Every error message states what happened and what the user should do next.		
NFR-UX-05	Interface meets WCAG 2.1 AA for contrast, focus visibility, and keyboard navigation on core flows.		
NFR-UX-06	Every list and empty state provides a next action rather than a blank panel.		
NFR-UX-07	Operator UI follows **Context-Rich, Not Clutter-Rich** (Brain Dump `# UX/UI Design Direction`): strong hierarchy, progressive disclosure, ~20% less visual clutter than typical dense dashboards; information density without visual density.		
NFR-UX-08	Primary merchant UI language is French for the Tunisian market; currency displayed in DT with locale formatting; AI conversation language is separate from UI language.		
NFR-UX-09	AI activity, consequential actions, and plan usage limits are visible in business language —not raw model metrics—for ordinary users.		
NFR-UX-10	Dashboard and overview screens show 4–6 relevant metrics per view, not every possible KPI; MVP analytics remain exactly four canonical KPIs (§15).		
NFR-UX-11	Visual design follows `20_DESIGN_SYSTEM.md` (OnchainStack lime as restrained accent, dark header, calm B2B aesthetic; latest generated dashboard is the visual benchmark).		

16.7 Observability and operability

ID	Requirement	Target or criterion	Verify
NFR-OPS-01	Structured JSON logging with correlation identifiers spanning webhook → job → outbound send.		
NFR-OPS-02	Metrics exported for queue depth, job failure rate, AI latency, tool success rate, provider error rate, and per-tenant message volume.		
NFR-OPS-02a	CloudWatch tracks API errors, worker failures, AI latency, and Meta API failures.		
NFR-OPS-03	Alerting on dead-letter queue growth, webhook signature failures, LLM error rate, and database saturation.		

ID	Requirement	Target or criterion	Verify
NFR-OPS-04	Health and readiness endpoints suitable for ECS.		
NFR-OPS-05	Database migrations are versioned, forward-only, and run automatically on deploy (after ECS rollout in CI).		
NFR-OPS-06	The full stack runs locally via a single Docker Compose command with seeded demo data.		
NFR-OPS-07	Production uses ECS Fargate, RDS, ElastiCache, S3, Secrets Manager, CloudWatch, and an ALB. No Kubernetes, no NAT Gateway in MVP, no extra AWS accounts. Canonical: 18_DEPLOYMENT.md.		

16.8 Maintainability and cost

ID	Requirement	Target or criterion	Verify
NFR-MNT-01	Module boundaries defined in 10_ARCHITECTURE.md are enforced; cross-module database access fails review.		
NFR-MNT-02	Shared domain types are defined once and reused by API and frontend.		
NFR-MNT-03	Automated tests cover every state transition, guardrail, and idempotency path; these are non-negotiable test targets regardless of aggregate coverage percentage.		
NFR-MNT-04	The AI golden set runs in CI on every prompt or model change and blocks merge on regression.		
NFR-COST-01	Average AI cost per handled conversation is tracked per business and must remain within the modelled margin for the plan.		
NFR-COST-02	Retrieval context size is bounded by configuration so a single conversation cannot consume unbounded tokens.		

Quality guardrails recorded with the product metrics

These targets are documented in the project overview and are release gates, not extra dashboard tiles. The measurement method lives in the AI architecture specification.

Metric	MVP target	Status
Answer accuracy on FAQ, price, and stock	Greater than 90% on a staff-reviewed sample	Documented target
Hallucination rate	Less than 2%	Documented target
Inbound to first AI reply	p95 under 30 seconds	Same commitment as NFR-PERF-02
Tool success rate	Greater than 99%	Documented target
Webhook acknowledgement	p95 under 500 ms	Same commitment as NFR-PERF-01

External constraints

- Meta messaging windows apply. A send outside the permitted window must surface a failure to staff.
- Instagram and Messenger permissions depend on Meta app review. Onboarding must tolerate a pending review.
- WhatsApp templates, the 24-hour window, and per-conversation billing are Phase 2.
- There is no payment provider. Every purchase path ends in human confirmation.

- LLM latency and availability are outside the product's control, so fallback, timeout, and escalation are required.
- Cash on delivery is the market norm. Payment status is set by staff, not by a gateway callback.
- The merchant is not assumed to be a developer. Onboarding must not require API keys or a custom spreadsheet.

Documented requirements and working assumptions

The following items are not additional acceptance criteria. They are recorded so a reader can see what is locked, what is deferred, and what is still a working answer.

Topic	Status	What the documents say
Modular monolith, official Meta APIs, human closes the sale, RLS off, three Fargate services, REST, pgvector, Socket.IO	Locked decision	ADR-001 through ADR-024. These are not proposals.
No Kubernetes, no NAT gateway, no Vercel, no second AWS account	Locked for MVP	ADR-024 and the deployment specification. Private tasks and multi-AZ RDS are explicitly "not now".
Field-level encryption of customer PII	Deferred shortcut	OQ-06 / OQ-71: volume encryption plus Secrets Manager for provider tokens. Field-level PII encryption is tech debt, not an MVP requirement.
Automatic 24-month purge of customer PII	Open	OQ-04 / OQ-72: targeted erasure on request is required (NFR-SEC-10). An automatic purge is not in MVP.
Currency charged for the SaaS plan	Open	OQ-02: the plan may be charged in TND or in the provider's EUR/USD. The merchant interface still shows DT. Catalog money is ISO TND.
Apex domain	Open	OQ-49. Host roles are fixed: app, api, and hooks. The domain name is not.
Merchant UI languages	Working answer, treated as frozen	OQ-01: fr-TN, then ar-TN with RTL, then en. Customer-facing AI language is per business.
Seven-day stale-handoff warning	Not in MVP	Conflict C-1: auto-close at 30 days with no pending order. The warning is Phase 2.
Stock after a reservation is committed	Specified, with a gap	Available stock subtracts only active, unexpired reservations. The reservation becomes committed when the pending order is created. A signed inventory movement is Should, not Must. This submission does not invent a further stock write.